Points 6 Published

Details Questions

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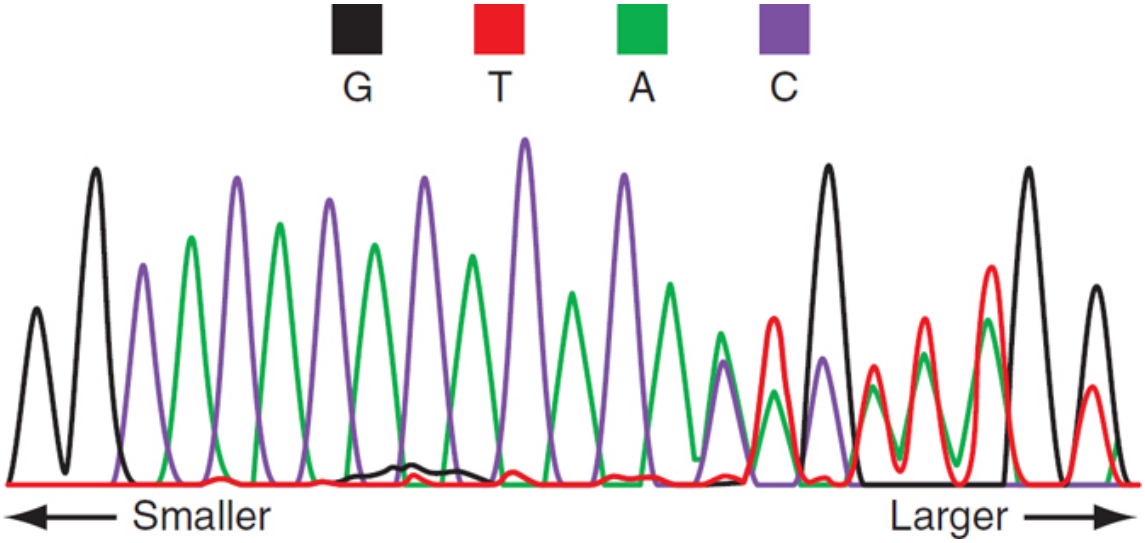
Question

Genomic DNA from a woman's blood cells is PCR amplified by a single pair of primers representing a unique locus in the genome. The PCR products are then sequenced by the Sanger method, using one of the PCR primers as a sequencing primer. The following figure shows a trace of just part of the sequence read.



Question 1.2 pts

What kind of polymorphism is most likely represented?

- SNP
- Indel
- SSR

Question 2.4 pts

With your answer to part (a) in mind, determine the woman's genotype at this locus. Select the two alleles. Nucleotides from each allele are given in their 5'-to-3' orientation.

- GGCACACACACACACACACACA
- GGCACACACACACAATGTTAGG
- GGCACACACACACAAGAAAGAT
- GGCACACACACACACACAATGT
- GGCACACACACACAGTATGGTT

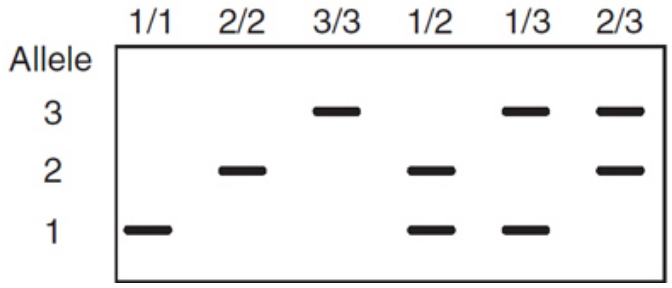
Question 1.2 pts

Another way to genotype this locus is to analyze the PCR products using gel electrophoresis.

- True
- False

Question 1.2 pts

Six diploid genotypes are present in this population.



The above is a gel of PCR products from the amplification of an SSR in a disease-associated gene in six different individuals. Allele 1 is dominant for the disease. If a man of genotype 1/1 and a female of genotype 1/2 were to have a child, what is the probability that the child would have the same PCR products as the 1/1 genotype showed above?

- 0%
- 25%
- 50%
- 66%
- 100%

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